Modulation part 2

Intro to Audio Tech 1
Jocelyn Kavanagh
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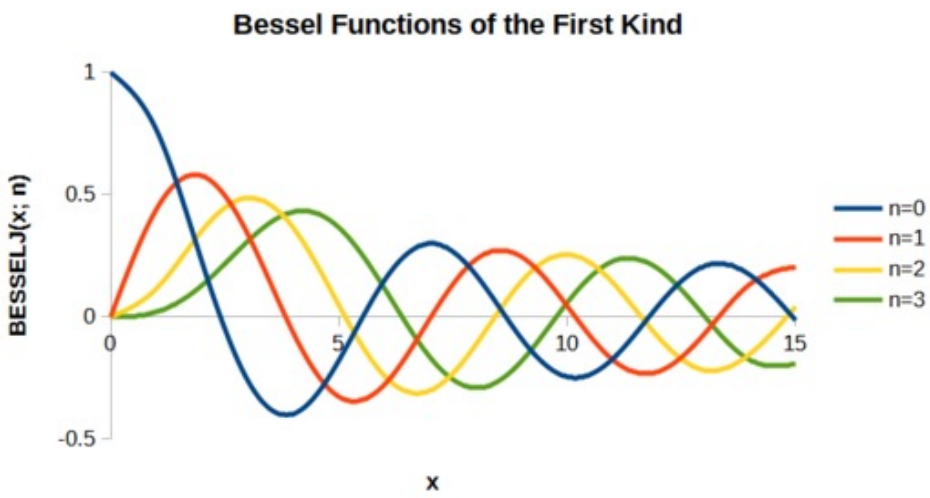
Lesson Outline

- Modulation Math
- Modulation in Max
- Up Next…

Modulation

Strength of Sidebands

- Sideband amplitudes are determined by Bessel Functions
  - Like dampened sinusoids
- Based on the order and index of modulation



Practice Questions

- Describe the frequency content of the output signal from the frequency modulation parameters below:

  Carrier: 400 Hz
  Modulator: 70 Hz
  Depth: 140

- Using ring modulation to modulate a 300 Hz square wave with a 100 Hz sinusoid, describe the frequency content assuming the square wave only has 5 harmonics (include the fundamental)

Modulation in Max

FM Synthesis

- To get FM to sound “good”, generally you want to control
  - the harmonic ratio (distance between frequencies) of the carrier and modulator
  - The depth of modulation (or the strength of the modulator) or the modulation index
  - Amplitude envelopes around the modulator(s)
- Common ratios
  - 1:1 – like AM, and with high modulator depth its more like a sawtooth
  - 2:1 – like a square wave, more so with high modulator depth
  - Higher ratios are used for plucking and bell sounds
  - Inharmonic ratios like 2:9, 2:7, 3:8 will create crunchier sounds

More on Frequency Ratios

C:M	Carrier Freq	Modulator Freq	GCF	Result Spectra
1:1	200	200	200	All harmonics with fundamental = carrier
1:2	200	400	200	Only odd harmonics with fundamental = carrier
N:1	1000	200	200	All harmonics with fundamental = modulator
Fm => 2*Fc	200	1340	40	Carrier is the lowest frequency (mirroring)
Fm < 2*Fc	200	150	50	Frequencies lower than carrier

- When you spell out all the partials created with the different integer ratios, you will find some share common harmonics
  - Ex. 1:2 and 3:8 share frequencies
- Barry Traux used these relationships to create Arras

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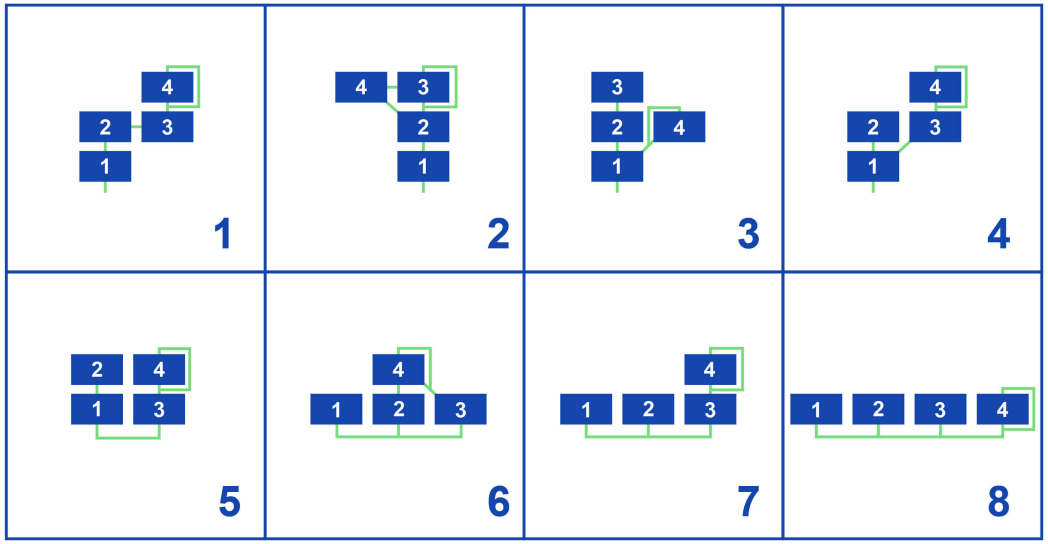
FM Synthesis Algorithms

- Most FM synths are not restricted to a singular carrier and modulator
- You can have multiple in different arrangements with varying envelopes and feedback parameters



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